

Research Methodology: Methods and Techniques (Exam-Oriented Notes)

Based on the Book by C.R. Kothari (Second Revised Edition) [2]

Chapter 1: Research Methodology: An Introduction

Meaning of Research

- **Definition:** A scientific and systematic search for pertinent information on a specific topic [20].
- **Concept:** It is an art of scientific investigation, a “systematized effort to gain new knowledge” [20], and an original contribution to the existing stock of knowledge making for its advancement [21].
- **Essence:** A movement from the known to the unknown; a voyage of discovery [20].

Objectives of Research

- **Core Purpose:** To discover answers to questions through the application of scientific procedures; to find out the truth which is hidden and has not been discovered as yet [23].
- **Classifications:**
 1. *Exploratory or Formulative:* Gain familiarity with a phenomenon or achieve new insights into it [23].
 2. *Descriptive:* Portray accurately the characteristics of a particular individual, situation, or group [24].
 3. *Diagnostic:* Determine the frequency with which something occurs or with which it is associated with something else [24].
 4. *Hypothesis-Testing:* Test a hypothesis of a causal relationship between variables [24].

Motivation in Research

- **Key Drivers:**
 - Desire to get a research degree along with its consequential benefits [24].
 - Desire to face the challenge in solving the unsolved problems [24].
 - Desire to get intellectual joy of doing some creative work [24].
 - Desire to be of service to society [24].
 - Desire to get respectability [24].

Types of Research

- **Descriptive vs. Analytical:**

- *Descriptive*: Surveys and fact-finding enquiries of different kinds. The researcher has no control over variables; they can only report what has happened or is happening (Ex-post facto research) [25].
- *Analytical*: The researcher has to use facts or information already available, and analyze these to make a critical evaluation [25].
- **Applied vs. Fundamental:**
 - *Applied*: Aims at finding a solution for an immediate problem facing a society or an industrial/business organisation [25].
 - *Fundamental (Basic/Pure)*: Mainly concerned with generalizations and with the formulation of a theory (“gathering knowledge for knowledge’s sake”) [25].
- **Quantitative vs. Qualitative:**
 - *Quantitative*: Based on the measurement of quantity or amount; applicable to phenomena that can be expressed in terms of quantity [26].
 - *Qualitative*: Concerned with qualitative phenomenon, i.e., involving quality or kind (e.g., Motivation Research to investigate reasons for human behavior using projective techniques) [26].
- **Conceptual vs. Empirical:**
 - *Conceptual*: Related to some abstract idea(s) or theory; generally used by philosophers and thinkers to develop new concepts or to reinterpret existing ones [27].
 - *Empirical (Experimental)*: Relies on experience or observation alone, often without due regard for system and theory. It is database-driven, coming up with conclusions which are capable of being verified by observation or experiment [27].
- **Other Types:**
 - *One-time vs. Longitudinal*: Confined to a single time-period vs. carried on over several time-periods [27].
 - *Field-setting vs. Laboratory vs. Simulation*: Based on the environment in which it is carried out [27].
 - *Clinical vs. Diagnostic*: Case-study or indepth approaches to reach basic causal relations [27].
 - *Exploratory vs. Formalized*: Development of hypotheses vs. substantial structure with specific hypotheses to be tested [27].
 - *Historical*: Utilizes historical sources like documents, remains, etc. to study events/ideas of the past [27].
 - *Conclusion-oriented vs. Decision-oriented*: Free to conceptualize/redesign as one proceeds vs. geared to the needs of a decision-maker [27].

Research Approaches

- **Two Basic Approaches:**
 1. *Quantitative*: Generation of data in quantitative form which can be subjected to rigorous quantitative analysis [28]. Sub-classified into:

- *Inferential*: Survey-based approach where a sample is studied to infer population characteristics [28].
 - *Experimental*: Greater control over the environment; some variables are manipulated to observe their effect [28].
 - *Simulation*: Construction of an artificial environment within which relevant information/data can be generated [28].
2. *Qualitative*: Subjective assessment of attitudes, opinions, and behavior. Uses techniques like focus group interviews, projective techniques, and depth interviews [26, 28].

Significance of Research

- **Progress & Inquiry**: “All progress is born of inquiry. Doubt is often better than overconfidence, for it leads to inquiry, and inquiry leads to invention” (Hudson Maxim) [29].
- **Inductive Thinking**: Inculcates scientific and inductive thinking and promotes logical habits of thinking [29].
- **Government Policy**: Serves as a vital tool for economic and social policy. Government research operates in three phases: (i) Investigation of economic structure, (ii) Diagnosis of events taking place, and (iii) Prognosis (prediction of future developments) [30].
- **Operational Problems**: Provides a quantitative basis for decisions in business and industry [29, 30].
- **Social Value**: Serves as a career path (degree), source of livelihood (professionals), outlet for new ideas (philosophers), developer of style (literary men), and generalizer of new theories (intellectuals) [31, 32].

Research Methods versus Methodology

- **Research Methods**: The methods/techniques used by the researcher to perform research operations (grouped as: data collection, statistical tools for establishing relationships, and methods to evaluate accuracy) [34, 35].
- **Research Methodology**: A way to systematically solve the research problem. It is the science of studying how research is done scientifically [36]. It includes not only the methods but also the logic behind them, their assumptions, and the criteria for choosing specific techniques [36].

Research and Scientific Method

- **Philosophy**: Scientific method is the pursuit of truth as determined by logical considerations [37].
- **Unity of Science**: “The unity of all sciences consists alone in its methods, not its material” (Karl Pearson) [37].
- **Aims**: To achieve a systematic interrelation of facts by experimentation, observation, and logical arguments from accepted postulates [37].

Importance of Knowing How Research is Done

- **Training:** Provides training in gathering material, card-indexing, field-work, statistics, questionnaires, and interpretation [40].
- **Objectivity:** Helps develop disciplined thinking or a ‘bent of mind’ to observe fields objectively [41].
- **Intelligent Consumption:** Enables the user to intelligently evaluate and judge the adequacy of research results and make rational decisions [42].

Research Process

- **Flow of Activities:**
 1. *Formulating the research problem* [44].
 2. *Extensive literature survey* [44].
 3. *Development of working hypotheses* [44].
 4. *Preparing the research design* [44].
 5. *Determining sample design* [44].
 6. *Collecting the data* [44].
 7. *Execution of the project* [44].
 8. *Analysis of data* [44].
 9. *Hypothesis testing* [44].
 10. *Generalisations and interpretation* [44].
 11. *Preparation of the report or presentation of the results* [44].
- *Note:* The steps overlap continuously rather than following a strict sequence [44].

Criteria of Good Research

- **Scientific Standards:**
 1. Purpose should be clearly defined and common concepts used [63].
 2. Research procedure should be described in sufficient detail to permit replication [63].
 3. Procedural design should be carefully planned to yield objective results [63].
 4. Flaws in design should be reported frankly along with estimates of their effects [63].
 5. Analysis of data should be adequate and methods of analysis appropriate; validity/reliability checked [64].
 6. Conclusions should be confined to those justified by data [64].
 7. Integrity and experience of the researcher warrant greater confidence [64].
- **Qualities:**
 - *Systematic:* Structured with specified steps [65].
 - *Logical:* Guided by logical reasoning (induction/deduction) [65].
 - *Empirical:* Related to concrete data from real situations [66].
 - *Replicable:* Allows results to be verified by repeating the study [66].

Problems Encountered by Researchers in India

- **Major Bottlenecks:**
 1. Lack of scientific training in research methodology [67].
 2. Insufficient interaction between university departments and business units [67].
 3. Sacrosanct concept of secrecy in business, making data collection difficult [67].
 4. Overlap and duplication of studies due to lack of information [68].
 5. Absence of a unified code of conduct for researchers [68].
 6. Difficulty of obtaining timely secretarial and computerial assistance [69].
 7. Unsatisfactory library management and delays in tracing resources [69].
 8. Delays in receiving governmental publications and raw statistics [70].
 9. Ambiguities in conceptualization and definitions [71].
-

Chapter 2: Defining the Research Problem

What is a Research Problem?

- **Definition:** A difficulty experienced by a researcher (individual/group) in a practical or theoretical situation, seeking a solution [74].
- **Components of a Problem:**
 1. An individual or a group experiencing some difficulty [77].
 2. Objective(s) to be attained [77].
 3. Alternative means (courses of action) for obtaining the objective(s) [77].
 4. Doubt in the mind of the researcher regarding the relative efficiency of alternatives [78].
 5. An environment to which the difficulty pertains [78].

Selecting the Problem

- **Guiding Principles:**
 - Avoid overdone subjects where throwing new light is difficult [80].
 - Controversial subjects should not be chosen by an average researcher [80].
 - Avoid too narrow or too vague problems [80].
 - The subject must be familiar, feasible, and within reach of resources [80].
 - Consider researcher's background, training, budget, time, and participant cooperation [81].
 - Precede selection with a preliminary feasibility study [82].

Necessity of Defining the Problem

- **Value:** “A problem clearly stated is a problem half solved” [83].
- **Function:** Limits scope, separates relevant from irrelevant data, and dictates the research design and analytical techniques to be used [83, 84].

Technique Involved in Defining a Problem

- **Sequential Steps:**
 1. *Statement of the problem in a general way:* Expressing the practical/intellectual interest with pilot survey if needed [86].
 2. *Understanding the nature of the problem:* Discussing its origin and objectives with those who raised it [87].
 3. *Surveying the available literature:* Reviewing conceptual and empirical studies to find theoretical gaps and narrow the focus [88].
 4. *Developing the ideas through discussions (experience survey):* Seeking advice from experienced colleagues to sharpen focus [89].
 5. *Rephrasing the research problem:* Defining technical terms, making basic assumptions explicit, and putting the problem in highly specific operational terms [90, 92].

An Illustration

- *General:* “Why is productivity in Japan so much higher than in India?” [93]
- *Narrowed:* “What factors were responsible for the higher labour productivity of Japan’s manufacturing industries during the decade 1971 to 1980 relative to India’s manufacturing industries?” [94]
- *Operational:* “To what extent did labour productivity in 1971 to 1980 in Japan exceed that of India in respect of 15 selected manufacturing industries? What factors were responsible for the productivity differentials between the two countries by industries?” [94]

Conclusion

- Defining a problem follows a sequential pattern of clarifying ambiguities, resulting in a realistic, analytically meaningful, and operationally viable proposition [96].

Chapter 3: Research Design

Meaning of Research Design

- **Definition:** The blueprint for the collection, measurement, and analysis of data; arrangement of conditions to combine relevance to research purpose with economy in procedure [99].

- **Parts of Research Design:**
 - *Sampling design:* Selection of items to be observed [101].
 - *Observational design:* Conditions under which observations are made [102].
 - *Statistical design:* Quantitative choices regarding sample size and analysis [102].
 - *Operational design:* Techniques to carry out specified procedures [102].

Need for Research Design

- **Function:** Facilitates smooth sailing of research operations, maximizing information while minimizing expenditures of time, effort, and money [104].

Features of a Good Design

- **Attributes:** Flexible, appropriate, efficient, and economical [105].
- **Standard:** Minimizes bias, maximizes reliability, and yields the smallest experimental error [105].

Important Concepts Relating to Research Design

- **Dependent & Independent Variables:**
 - *Variable:* A concept which can take on different quantitative values [107].
 - *Continuous Variable:* Can assume any numerical value within a range (e.g., age) [108].
 - *Discrete Variable:* Expressed only in integers with distinct gaps (e.g., number of children) [108].
 - *Dependent vs. Independent:* If variable Y is a consequence of variable X , Y is dependent and X is independent [108].
- **Extraneous Variable:** Independent variables not related to the study's purpose but affecting the dependent variable (causing experimental error) [108].
- **Control:** Minimizing the influence of extraneous variables using design restraints [108].
- **Confounded Relationship:** When the dependent variable's change is not free from the influence of extraneous variables [108].
- **Research Hypothesis:** A predictive statement that relates an independent variable to a dependent variable to be tested [109].
- **Experimental and Non-experimental Hypothesis-Testing Research:**
 - *Experimental:* The independent variable is manipulated [109].
 - *Non-experimental:* The independent variable is NOT manipulated (e.g., correlation of IQ and reading ability) [109].

- **Experimental & Control Groups:** Exposed to special/novel conditions (experimental) vs. usual/baseline conditions (control) [110].
- **Treatments:** The different conditions under which groups are placed [110].
- **Experiment:** Process of examining the truth of a statistical hypothesis (absolute or comparative) [110].
- **Experimental Unit(s):** Pre-determined plots or blocks where treatments are applied [111].

Different Research Designs

- **Exploratory (Formulative) Studies:** Focuses on discovering ideas and insights. Design must be flexible [112]. Three main methods:
 1. *Survey of concerning literature:* Reviewing existing theoretical/empirical works [113].
 2. *Experience survey:* Interviewing competent people with practical experience [114].
 3. *Analysis of ‘insight-stimulating’ examples:* Intensive study of selected striking instances [115].
- **Descriptive and Diagnostic Studies:**
 - *Descriptive:* Portrays characteristics of individuals, groups, or situations [117].
 - *Diagnostic:* Determines the frequency of something or its association with something else [117].
 - *Design:* Must be rigid (not flexible) to protect against bias and maximize reliability. Relies on structured instruments and pre-planned statistical computations [117, 119, 120].
- **Hypothesis-Testing Studies:** Experimental designs permitting causal inferences while minimizing bias [106].

Basic Principles of Experimental Designs

- **Fisher’s Three Principles:**
 1. *Principle of Replication:* Repeating the experiment more than once to apply treatments to multiple units, thereby increasing statistical accuracy and precision [123].
 2. *Principle of Randomization:* Randomly assigning treatments to experimental units to safeguard against extraneous variables and assure objective probability measurements [124].
 3. *Principle of Local Control:* Dividing units into homogeneous blocks and varying extraneous factors deliberately to measure and eliminate soil/block variability from experimental error [124].

Conclusion

- The selection of design must be decided in advance based on the universe type, study objective, sampling frame, and desired accuracy [137].

Appendix: Developing a Research Plan

- **Research Plan Items:** Clear statement of objectives, explicit description of the problem, operational definitions of concepts, method/approach to be used, detailed technical procedures (quantification details), population to be studied (sampling frame), methods of data processing/analysis, results of pilot tests, and time/cost budgets [141, 142, 143, 144].
-

Chapter 4: Sampling Design

Census and Sample Survey

- **Census:** Complete enumeration of all items in the population. It is expensive, time-consuming, prone to non-sampling bias, and often physically impossible [145].
- **Sample Survey:** Examining a representative subset (sample) of the population. Yields sufficiently accurate results with less cost, time, and energy [145].

Implications of a Sample Design

- **Sampling Design:** A definite plan determined before data collection to obtain a sample from a given population [127].

Steps in Sampling Design

1. *Type of Universe:* Define the set of objects to be studied (Finite vs. Infinite) [146].
2. *Sampling Unit:* Choose geographic (district), social (family), construction (house), or individual units [147].
3. *Source List (Sampling Frame):* Prepare/acquire a reliable, comprehensive, and representative list of items of the universe [147].
4. *Size of Sample:* Choose an optimum sample size ensuring efficiency, representativeness, reliability, and cost-effectiveness [148].
5. *Parameters of Interest:* Determine whether estimating population mean, variance, or proportion [148].
6. *Budgetary Constraint:* Account for funds available for sampling [148].
7. *Sampling Procedure:* Decide the actual type of sample selection [127].

Criteria of Selecting a Sampling Procedure

- **Errors in Sampling:** Minimize systematic bias (caused by inappropriate frame, non-response, measurement errors) and sampling errors (random fluctuations) [128, 141].

Characteristics of a Good Sample Design

- Must result in a truly representative sample, have small sampling error, be economically viable, control systematic bias, and allow generalization [128].

Different Types of Sample Designs

- **Probability vs. Non-probability Sampling:**
 - *Non-probability*: Deliberate selection based on convenience or judgement. Probability of selection is unknown (Haphazard, convenience, purposive, quota, judgement sampling) [59].
 - *Probability (Random/Chance)*: Every item has an equal and known chance of selection (Simple random sampling, complex random sampling) [149].

How to Select a Random Sample?

- **Lottery Method**: Draw folded slips from a container [149].
- **Random Number Tables**: Use Tippet's or other standardized tables to read numbers sequentially [150].

Random Sample from an Infinite Universe

- Drawn by maintaining control over selection processes where each drawn item does not affect the infinite pool's properties [11].

Complex Random Sampling Designs

- **Systematic Sampling**: Selecting every k^{th} item after a random start [130].
- **Stratified Sampling**: Dividing a heterogeneous population into homogeneous, non-overlapping subpopulations (strata) and drawing a random sample from each stratum [155].
 - *Proportional*: Strata sample size proportional to strata population size [132].
 - *Optimum (Disproportionate)*: Strata sample size based on standard deviation and cost ($n_i \propto N_i \sigma_i$) [152].
- **Cluster & Area Sampling**: Dividing population into clusters, randomly selecting clusters, and completely enumerating all units within chosen clusters [137].
- **Multi-stage Sampling**: Sampling in successive stages (e.g., district $\rightarrow$ town $\rightarrow$ household) [138].
- **Sequential Sampling**: The sample size is not fixed in advance; sampling continues based on cumulative intermediate results [140].

Chapter 5: Measurement and Scaling Techniques

Measurement in Research

- **Definition:** The process of assigning numbers to physical objects or abstract concepts according to rules of correspondence mapping a domain onto a range [155, 156].

Measurement Scales

- **Four Types:**
 1. *Nominal:* Categorical classification with no quantitative value or order (e.g., Gender: 0 for male, 1 for female) [156].
 2. *Ordinal:* Ranks items in order of magnitude (relative positions) but distances between positions are unequal or unknown (e.g., military ranks) [152].
 3. *Interval:* Ranks with equal, standardized intervals between positions but has an arbitrary zero point (e.g., Fahrenheit temperature) [152].
 4. *Ratio:* Ranks with equal intervals and an absolute, true zero point representing absence of the trait (e.g., weight, height) [152].

Sources of Error in Measurement

- **Four Factors:**
 1. *Respondent:* Fatigue, boredom, anxiety, or ignorance [161].
 2. *Situation:* Presence of others, noise, or lack of privacy [161].
 3. *Measurer:* Interviewer bias, rewording, or improper coding [120, 161].
 4. *Instrument:* Defective layout, ambiguous wording, or formatting issues [162].

Tests of Sound Measurement

- **Three Key Criteria:**
 1. *Validity:* Extent to which an instrument measures what it claims to measure [157].
 - *Content:* Adequate coverage of the topic (judgmental/intuitive) [158].
 - *Criterion-Related (Predictive/Concurrent):* Ability to predict or relate to an external criterion [158, 159].
 - *Construct:* Extent to which it measures a theoretical trait [158].
 2. *Reliability:* Stability (consistent over time) and Equivalence (consistent across items/investigators) [160].
 3. *Practicality:* Judged by economy (budget trade-offs), convenience (easy layout), and interpretability (supplemented by manual and scoring keys) [162].

Technique of Developing Measurement Tools

- **Four Stages:** (a) Concept development, (b) Specification of concept dimensions, (c) Selection of indicators, and (d) Formation of an index [163].

Scaling

- **Definition:** Procedures of assigning numbers to degrees of opinions, attitudes, or institutional concepts along a continuum [167].

Scale Classification Bases

- Classified by: (a) Subject orientation, (b) Response form, (c) Degree of subjectivity, (d) Scale properties, (e) Number of dimensions, and (f) Scale construction techniques [168].

Important Scaling Techniques

- **Rating Scales:** Absolute evaluation against categories without reference to other objects (e.g., graphic scale or itemized scale) [171].
- **Ranking Scales:** Relative evaluation comparing objects directly (e.g., Paired Comparison, Method of Equal-Appearing Intervals, or Rank Order) [171, 172].

Scale Construction Techniques

- **Five Major Approaches:**
 1. *Arbitrary:* Ad-hoc, subjective item selection [169].
 2. *Consensus (Thurstone):* Submitted to judges who sort statements into 11 piles. Statements with high agreement (low variance) and evenly spread median values constitute the final scale [177, 178].
 3. *Item Analysis (Likert Summated):* 5-point agreements scored. Extreme top and bottom 25% groups act as criteria; items with high discriminatory power are retained [179, 180, 181, 182].
 4. *Cumulative (Guttman Scalogram):* Unidimensional, structured series where agreement with an extreme item implies agreement with all less extreme items. Evaluated by Coefficient of Reproducibility ($C_R = 1 - e/(nN)$; minimum standard 0.90) [185, 186, 191].
 5. *Factor Scales:* Based on multidimensional intercorrelations [192].
 - *Semantic Differential:* Charles Osgood’s 7-point bipolar ratings evaluating three dominant dimensions: Evaluation, Potency, and Activity [193, 194].
 - *Multidimensional Scaling (MDS):* Plots judged similarities as physical distances in a multidimensional geometric space [198].

Chapter 6: Methods of Data Collection

Collection of Primary Data

- **Primary Data:** Collected afresh and for the first time; original in character [203].

Observation Method

- **Definition:** Systematic viewing of raw behavior [96].
- **Types:** Structured vs. Unstructured [96], Participant vs. Non-participant [96], and Controlled vs. Uncontrolled [97].
- **Evaluation:** Eliminates subjective bias and record independent of respondent willingness, but is expensive and limited to current behavior [96, 97].

Interview Method

- **Personal Interview:** Structured (standardized set of questions) or Unstructured (flexible sequence/wording) [204].
- **Other Types:** Focussed (focuses on concrete experiences), Clinical (broad motivations), Non-directive (interviewer acts as a passive catalyst), and Telephone [204, 205].
- **Evaluation:** High response rate, deeper probing, but expensive and prone to interviewer bias [203].

Collection of Data through Questionnaires

- **Definition:** Mailed or sent series of typed questions to be completed independently by respondents [208].
- **Main Aspects:** General form (structured vs. unstructured) [209], Question sequence (general-to-specific; vital questions in middle) [210, 211], and Question formulation (simple, concrete, unambiguous, avoiding “danger/catch” words) [211, 213].
- **Wording Forms:** Multiple choice (closed) vs. Open-ended (latitude in phrasing) [212].
- **Essentials:** Short, attractive physical appearance, clear instructions, and includes control questions for verification [214, 215].

Collection of Data through Schedules

- **Definition:** Questionnaires filled out in person by trained enumerators [104]. Enumerators must be honest, patient, and trained to cross-examine [216].

Difference between Questionnaires and Schedules

- *Questionnaires:* Cheaper, sent by mail, relies on respondent literacy, suffers from high non-response, and has no direct contact [214].

- *Schedules*: Costly, filled out by enumerators, can use illiterate respondents, low non-response, and permits identity verification [214, 216].

Some Other Methods of Data Collection

- **Pantry & Store Audits**: Physical count of stocks [106].
- **Consumer Panels**: Transcribe continuous diaries of purchases [106].
- **Projective Techniques**: Indirect assessment of unconscious motivations [108].
 - *Word Association & Sentence/Story Completion* [217, 218].
 - *TAT & HIT (Holtzman Inkblot Test)*: HIT uses 45 cards with single responses to assess barrier scores [219, 221].
 - *Play & Sociometry (sociological patterns)* [222].
- **Depth Interviews & Content Analysis**: Motivational probing vs. systematic qualitative evaluation of documents [223, 224].

Collection of Secondary Data

- **Secondary Data**: Already collected and analyzed by someone else [226].
- **Criteria for Use**:
 1. *Reliability*: Who collected it? What methods? Was there bias? [227]
 2. *Suitability*: Do definitions and scope match the present study? [227]
 3. *Adequacy*: Accuracy levels and physical coverage boundaries [228].

Selection of Appropriate Method for Data Collection

- Depends on the: (i) Nature, scope, and object of enquiry, (ii) Availability of funds, (iii) Time factor, and (iv) Precision required [229].

Case Study Method

- **Definition**: Intensive qualitative observation and in-depth analysis of a single social unit (person, family, institution, community) as an integrated totality [230, 231].
- **Phases**: Locating the status, collecting history, diagnosing causal factors, applying therapy, and follow-up [234, 235].
- **Assumptions**: Uniformity in basic human nature; completeness of historical sequence [234].

Chapter 7: Processing and Analysis of Data

Processing Operations

- **Editing**: Scrutinizing completed schedules/questionnaires to detect and correct errors and omissions.

- *Field Editing*: Completed by investigator immediately on same/next day [252].
- *Central Editing*: Thorough, centralized check by a single editor/team [253].
- **Coding**: Assigning numbers/symbols to categories of answers. Categories must be exhaustive, mutually exclusive, and unidimensional [255, 256].
- **Classification**: Dividing data into homogeneous groups.
 - *By Attributes (Descriptive)*: Qualitative presence/absence (simple or manifold classification) [258].
 - *By Class Intervals (Numerical)*: Quantitative measures. Size of interval (i) determined using Sturges' Formula ($i = R/(1 + 3.3 \log N)$) [259, 262]. Limits can be *exclusive* (upper limit excluded) [264] or *inclusive* (upper limit included) [265].
- **Tabulation**: Orderly arrangement of summarized data in columns and rows [267]. Requires a distinct table number, clear title, captions (column headings), stubs (row headings), units, and footnotes [269, 270].

Some Problems in Processing

- **Don't Know (DK) Responses**: Solved by design adjustments, allocation from other data, or proportionate distribution [274, 275].
- **Percentages**: Reducing data to a 0-100 base. Never average separate percentages unless weighted by group size; higher figures are base for decreases [275, 276].

Elements/Types of Analysis

- **Descriptive**: Largely the study of distributions of one variable (unidimensional), two variables (bivariate), or more (multivariate) [277].
- **Inferential (Statistical)**: Testing hypotheses to determine validity and make generalizations [279].

Statistics in Research

- **Descriptive Statistics**: Development of indices from raw data [279].
- **Inferential Statistics**: Generalization from sample to population parameters [279].

Measures of Central Tendency

- **Arithmetic Mean**: Sum of values divided by n (simple or weighted; stable, algebraic, but affected by extreme items) [280, 281].
- **Median**: Positional middle value of an arrayed series (unaffected by extremes) [285].
- **Mode**: Most frequently occurring value; concentration peak (useful for popular sizes) [282].

- **Geometric & Harmonic Means:** Reciprocals of arithmetic averages of logarithms and reciprocals [133, 283].

Measures of Dispersion

- **Range:** Difference between largest and smallest item [134].
- **Mean Deviation:** Average of absolute deviations from mean/median/mode [284].
- **Standard Deviation (σ):** Root-mean-square deviation. Gives the measure of dispersion and basis for standard error [135, 303].
- **Coefficient of Variation:** Ratio of standard deviation to mean multiplied by 100 [136].

Measures of Asymmetry (Skewness)

- Indicates shape of distribution and divergence from normal symmetry [136].

Measures of Relationship

- **Simple Correlation (Karl Pearson's r):** Linear relationship magnitude (-1 to +1) [139, 298].
- **Partial & Multiple Correlation:** Relationship of variables when others are controlled or combined [142, 143].

Simple Regression Analysis

- Formulating a mathematical model ($Y = a + bX$) for predicting a dependent variable based on independent values [285].

Multiple Correlation and Regression

- Simultaneous assessment of multiple independent variables on a single metric dependent variable [142, 143].

Association in Case of Attributes

- Tested via Yule's Coefficient of Association (Q) [286]. Illusory association occurs when real relationship is absent [287]. Tested on $r \times c$ contingency tables [288].

Other Measures

- **Index Numbers:** Approximate indicators of directional trend change over time [290].
- **Time Series Analysis:** Decomposing data into Trend (T), Cyclical (C), Seasonal (S), and Irregular (I) components [291].

Chapter 8: Sampling Fundamentals

Need for Sampling

- Enables rapid results, reduces costs, controls non-sampling errors, and permits destructive testing [152, 300].

Some Fundamental Definitions

- **Universe/Population:** Total items in field of inquiry (elementary units) [300].
- **Statistic vs. Parameter:** Sample-derived indices vs. Population-derived values [311].
- **Sampling Frame:** Source list of population items [311].
- **Sampling Error:** Fluctuations due to random chance [311].

Important Sampling Distributions

- Normal, Student's t , Chi-square, and F distributions, depending on parameters and sample sizes [155, 311].

Central Limit Theorem

- Sample means of samples drawn from *any* population approach normal distribution as sample size n increases ($n > 30$) [301].

Sampling Theory

- Deals with math of sampling accuracy and standard error based on laws of statistical regularity and inertia of large numbers [158].

Sandler's A-test

- A paired-sample shortcut alternative to Student's t -test based on difference scores: $A = \sum D_i^2 / (\sum D_i)^2$ [302].

Concept of Standard Error

- The standard error is the standard deviation of the sampling distribution. Smaller S.E. indicates greater reliability [303].

Estimation

- **Point Estimate:** A single value estimate [311].
- **Interval Estimate:** A range of values within confidence limits [311].

Estimating the Population Mean (μ)

- Calculated as $\bar{X} \pm z \cdot S.E.$ using finite population multipliers if needed [305, 336].

Estimating Population Proportion

- Calculated as $p \pm z \cdot \sqrt{pq/n}$ for attribute frequencies [172].

Sample Size and its Determination

- Depends on: Universe nature (heterogeneity), number of proposed classes, precision, and confidence level [305, 306].

Determination of Sample Size: Precision & Confidence Approach

- **Means:** $n = z^2 \sigma_p^2 / e^2$ [307].
- **Proportions:** $n = z^2 pq / e^2$ [308].

Determination of Sample Size: Bayesian Approach

- Maximizes Expected Net Gain (ENG) by balancing Expected Value of Sample Information (EVSI) against cost of sampling [309].
-

Chapter 9: Testing of Hypotheses-I (Parametric Tests)

What is a Hypothesis?

- **Definition:** A tentative predictive statement or assumption made to test logical or empirical consequences [317].
- **Characteristics:** Clear, testable, specific, simple, limited in scope, and consistent with known facts [318, 319, 320].

Basic Concepts Concerning Testing of Hypotheses

- **Null (H_0) vs. Alternative (H_a):** Hypothesis of no difference vs. the operational alternative to prove [321, 323].
- **Level of Significance (α):** Maximum probability of Type I error (rejecting true H_0) chosen in advance [324, 325].
- **Type I vs. Type II Errors:**
 - *Type I (α):* Rejecting true H_0 [327].
 - *Type II (β):* Accepting false H_0 [327].
- **Two-tailed vs. One-tailed Tests:** Rejection region on both tails ($H_a : \mu \neq \mu_0$) vs. single tail ($H_a : \mu > \mu_0$) [328].

Procedure for Hypothesis Testing

- **Sequence:** Formulate $H_0/H_a \rightarrow$ specify α *ightarrow* select sampling distribution *ightarrow* calculate test statistic *ightarrow* find probability *ightarrow* accept or reject H_0 [331].

Important Parametric Tests

- **z-test:** $n > 30$ or known population variance [333, 334].
- **t-test:** $n \leq 30$ and unknown population variance [335].
- **F-test:** Compares the variance of two normal populations [333].
- **Sandler's A-test:** Shortcut pairing test [335].

Limitations of the Tests of Hypotheses

- Tests are not absolute proofs; they are tools of decision-making subject to probability limitations and require judgmental context [358].
-

Chapter 10: Chi-Square Test

Chi-Square as a Test for Comparing Variance

- Tests whether sample variance matches a hypothesized population variance: $\chi^2 = \sigma_s^2(n-1)/\sigma_p^2$ [16, 367].

Chi-Square as a Non-parametric Test

- **Goodness of Fit:** Compares observed (O) and expected (E) frequencies: $\chi^2 = \sum (O_i - E_i)^2 / E_i$ [301].
- **Independence:** Evaluates whether two attributes in a contingency table are associated [333].

Conditions for the Application of χ^2 Test

1. Observations randomly collected [368].
2. Items in sample independent [368].
3. Expected frequencies in any cell must be ≥ 10 (combine adjacent cells if smaller; some accept ≥ 5) [368].
4. Total $N \geq 50$ [369].

Yates' Correction

- Applied to 2×2 tables with expected cell frequencies < 5 by subtracting 0.5 from absolute differences: $\chi_{adj}^2 = \sum (|O - E| - 0.5)^2 / E$ [16, 655].
-

Chapter 11: Analysis of Variance and Covariance

Analysis of Variance (ANOVA)

- **Definition:** Decomposes total variance into component parts attributable to specific causes vs. random error [361, 364].

The Basic Principle of ANOVA

- Compares variance between treatment groups against variance within groups using the F -ratio: $F = MS_{between}/MS_{within}$ [17, 381].

ANOVA Technique

- **One-way ANOVA:** Compares k sample means [382].
- **Two-way ANOVA:** Compares treatment effects and controls a metric covariate, evaluating interaction effects [127, 392].
- **ANOVA in Latin-Square Design:** Controls two extraneous sources of variation (e.g., soil fertility rows and seed columns) [129, 398].

Analysis of Co-variance (ANOCOVA)

- Controls the influence of an uncontrolled metric independent variable (co-variate) on the dependent variable using linear regression [391, 402].
-

Chapter 12: Testing of Hypotheses-II (Nonparametric Tests)

Important Nonparametric or Distribution-free Tests

- Do not rely on population normality assumptions [417].
- **Sign Test:** One-sample or two-sample matched pair evaluations based on signs of differences [418, 419].
- **McNemer Test:** Evaluates significance of change in nominal matching before-and-after views [423].
- **Wilcoxon Signed Rank Test:** Ranks absolute differences to check matched pairs [615].
- **Mann-Whitney U-test:** Compares medians of two independent groups using rank sums [665].
- **Kruskal-Wallis H-test:** Nonparametric one-way ANOVA utilizing rank sums [425, 427].
- **Runs Test:** Evaluates sequence randomness based on number of runs of binary attributes (r) [428].
- **Spearman Rank Correlation (r_s):** Measures ordinal association [430].
- **Kendall Concordance (W):** Measures consensus agreement among k judges ranking N items [433].

Chapter 13: Multivariate Analysis Techniques

Classification of Multivariate Techniques

- **Dependence:** Explanatory variables predict criterion variables.
 - *Multiple Regression:* Metric criterion predicted from metric explanatory variables [450].
 - *Discriminant Analysis:* Non-metric categorical group predicted from metric variables [452, 456].
 - *MANOVA:* Multiple metric dependent variables evaluated against non-metric experimental factors [456].
 - *Canonical Correlation:* Predicts a metric set from another metric set [456].
- **Interdependence:** Evaluates mutual relationships without dependency.
 - *Factor Analysis:* Groups correlated manifest variables into fewer latent factors [457, 458].
 - *Cluster Analysis:* Classifies entities into homogeneous, distinct groups [496].
 - *MDS:* Maps metric/non-metric perceptual similarities into geometric space [198, 500].
- **Path Analysis:** Uses path diagrams to decompose total correlation into direct and indirect effects using standardized regression beta weights [501, 504].

Chapter 14: Interpretation and Report Writing

Meaning and Necessity of Interpretation

- Refers to drawing inferences from collected facts; links findings to existing theories and establishes continuity [507, 508].

Different Steps in Writing Report

1. Logical analysis of subject-matter (logical/chronological) [518].
2. Preparation of final outline [518].
3. Preparation of rough draft [519].
4. Rewriting and polishing [519].
5. Preparation of final bibliography (books and periodicals list) [519, 520].
6. Writing the final draft [522].

Layout of the Research Report

- **Preliminary Pages:** Title, preface/foreword, Table of Contents, list of tables/charts [524].

- **Main Text:** Introduction, statement of findings/recommendations, results, implications, and summary [524].
- **End Matter:** Appendices, bibliography, and alphabetical subject/author index [530].

Types of Reports

- **Technical Report:** Focuses on methods, assumptions, detailed data, and limitations [533, 534].
- **Popular Report:** Focuses on practical findings, simplicity, subheadings, and visual charts [537].

Chapter 15: The Computer: Its Role in Research

The Computer System

- **CPU Segments:** Internal Storage (holds programs/data), Control Unit (sequencing), and Arithmetic Logical Unit (ALU - calculations) [570].
- **Terminology:** Hardware, Software (programs), Firmware, and System/Application Software [573, 574].

Important Characteristics

- High speed, diligence (no fatigue), storage, accuracy, and automation [575, 576, 577].

The Binary Number System

- Operates on base-2 (0 and 1). Arithmetic is reduced to rapid binary addition using complement subtraction [570, 581, 582].

Computers and Researcher

- Enables rapid statistical analysis (Canned packages for regression, factor analysis, ANOVA) [587, 588].
- *Limitation:* **GIGO** (“Garbage In, Garbage Out”) – the quality of the analysis is entirely dependent on the quality of inputs and programming [592, 593].